

Literature Replication

Perspectives on AI, Business and Sustainability (PAIBS)

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Term 1 (October 8th - December 3rd) → *From challenge to solution concept*

Definition and Solution Design - identifying meaningful challenges, understanding stakeholder needs, and proposing comprehensive solution.

Term 2 (February 12nd - May 14th) → *From concept to working prototype*

Technical Implementation - technical development, culminating in a fully developed prototype and business case.

Term 2		
Date	Content	Teacher/s
February 12, 2026	Literature Replication + Recap	Marc Herrera
February 19, 2026	Prototyping tools and techniques	Jose Rodriguez
March 12, 2026	Front-end prototyping	Jose Rodriguez
April 9, 2026	Prototyping AI applications	Jose Rodriguez
April 16, 2026	Interpreting AI-Based EO Results and Value	Marc Herrera
April 30, 2026	Impact assessment	Angelo Romasanta
May 14, 2026	Final Presentations	Marc Herrera / Angelo Romasanta

Introduction: Grading & Final Deliverables

Task	Grade Weight	Description / Sub-grading
Group project work & presentations	60%	- Problem definition & research quality (20%) - Technical implementation (50%) - Business validation (20%) - Communication & presentation (10%). Individual grade adjusted by peer-review multiplier (x0.8–x1.2).
Midterm individual deliverable	15%	Interview report (user research) and technical literature critique related to the group project.
Final exam	25%	Individual exam covering technical and methodological content from both terms.

Team Projects Discussion - Activity

Status - What are we doing now?

- Data Preparation
- Data Processing
- AI Model Training
- Results Interpretation

Main Problems - What problems are we facing?

- Data Obtention
- Model Performance
- Insights Interpretation

Questions - What doubts do we have?

- ?

Definition:

- Not just a summary, but a systematic synthesis of existing knowledge.

The Goal:

- To identify the "**Gap**" — what do we know, and what is missing?

The Funnel Method:

- Moving from a broad field to your specific research question.

Key Output:

- Building a theoretical framework that justifies your own study.

What Is Literature Replication?

Repeating a published study to **test reliability of findings**. Uses the **same or similar methods as the original**.

Checks whether results hold:

- Over time
- Across samples
- In different contexts

Why it **matters**

- Strengthens scientific credibility
- Detects errors, bias, or overfitting
- Supports cumulative knowledge



"Brilliant! But just so we're clear... this was my idea if it works out... yours if it doesn't."

Published findings fail to yield the same results when repeated, challenging the reliability of research.

1. Direct Replication

Using the exact same methods, materials, and conditions as the original study.

2. Conceptual Replication

Testing the same hypothesis but using different methods or a different population.

3. Partial / Analytical Replication:

Re-analyzing original data but testing alternative models or assumptions.

Internal vs. External:

- **Internal:** Same researchers, new data.
- **External:** Different researchers, new data.

Anatomy of a Research Paper/Article

Abstract - Provides a brief overview of the entire study to help decide if the paper is relevant for you.

Introduction - Establishes the general research problem and states the specific questions the study aims to answer.

Literature Review - Surveys existing research to show what is already known and identifies missing information to justify the new study.

Context - Describes the specific environment, time period, or population in which the research takes place.

Methods - Details the exact data collection and analysis techniques used so that other researchers can repeat the study.

Results - Presents the objective data, statistics, and observations derived from the methodology.

Discussion/Conclusion - Explains the significance of the results and admits any weaknesses or limitations in the study design.

Future Work - Proposes specific follow-up studies to address unanswered questions or overcome the current study's limitations.

Key Questions to Ask When Reading a Paper

How – Can I reproduce the method?

→ Are all steps, data, and decisions explicitly described?

Who – Is the sample appropriate and representative?

→ Who is *excluded* or underrepresented?

Bias – Who funded it? Any conflicts?

→ Who benefits if the results are true?

Leap – Do conclusions follow from results?

→ Where does interpretation begin to replace evidence?

Why – Is the research gap meaningful?

→ Helps students detect novelty inflation.

What if – Are assumptions and alternatives addressed?

→ Trains theoretical skepticism, not cynicism.

Reuse – Can this study be replicated or extended?

→ Bridges reading → doing (great for replication-focused courses).

Most Useful Sources

Source	Link	Best Used For...
Google Scholar	scholar.google.com	Quick, broad searches across all disciplines.
JSTOR	jstor.org	Deep archives in Humanities and Social Sciences.
EBSCOhost	search.ebscohost.com	Specialized databases (often accessed via university library).
ResearchGate	researchgate.net	Networking with authors and finding "gray literature."
Web of Science / Scopus	scopus.com	Citation tracking & systematic reviews
Open Science Framework	osf.io	Replication, open data, preregistration

Example

1. Tell me one **topic**.
2. Let's look at the latest literature. **What are we looking for?**
3. Analysis → **Is this useful for me/my project?**
 - a. Abstract
 - b. Introduction - Conclusions
 - c. Results
 - d. Rest
4. **Conclusions** - Key Insights
5. Why manual and not go **straight to Chat (AI)?**

- **The "Hallucination" Problem:** AI is a probability engine, not a database. It can confidently invent:
 - Fake Citations: Creating "**phantom**" **papers** with real-sounding titles and real author names. [Example](#)
 - Ghost DOIs: Providing **links** or identifiers that **lead nowhere** or to **unrelated studies**.
- **Logical "Fuzziness":** LLMs are excellent at mimicry but struggle with nuanced critical analysis or weighing **contradictory evidence**.
- **The Reproducibility Crisis:** Because AI output is stochastic (randomized), two researchers using the **same prompt may get different summaries**, making the process **non-replicable**.
- **Bias Reinforcement:** AI often mirrors the biases found in its training data, potentially skewing your review toward "**popular**" rather than "**accurate**" findings.

*If you use AI to summarize a paper, you must **verify every claim against the original document**. A single fake citation can result in a charge of **academic fraud**.*

Activity: Literature Analysis

With your group projects:

Step 1 Selection

- Choose a **topic** you like: health, fashion, sustainability, technology, ...
- Narrow it to a **specific** area: heart diseases (risk models), fast fashion (consumer behaviour), ...

Step 2 Look at the **latest literature** and identify **3–5 recent and influential papers**.

- 1 highly cited paper
- 1 very recent or “innovative” paper

Step 3 Answer → Synthesize

- a. What are the common findings across studies?
- b. Where do studies agree? Where do they diverge?
- c. Are conclusions incremental or genuinely disruptive?

Activity: Literature Analysis

Step 4 Replication

- Could this study be replicated?
- What type of replication would be most appropriate?
 - ◆ Direct
 - ◆ Conceptual
 - ◆ Analytical
- What would you keep the same? What would you change?

Step 5 Report

3-5 minutes group summary

- **One insight** from the literature
- **One weakness** or open question
- **One concrete replication idea**